CHAPTER 12

ICT AND SOCIETY

CHAPTER ESSENCE AT A GLANCE:
Evaluate ethical, legal, and environmental dimensions of ICT: intellectual property, cyber law, data privacy, e-waste lifecycle, and green computing.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Digital Governance and Public Services:** Electronic governance harnesses information communication technologies to deliver transparent, accountable, and efficient administrative services to citizens, commercial enterprises, and government departments.

• **Intellectual Property Protection:** Intellectual property legal frameworks grant creators exclusive legal rights over original creations, distinguishing between copyright for artistic and literary expressions, patents for novel technological inventions, and trademarks for distinctive commercial symbols.

• **Cyber Crimes and Digital Threats:** Digital communication networks face various unauthorized activities including deceptive phishing schemes to steal user credentials, illicit server intrusions, distributed denial of service outages, and disruptive malicious software.

• **Layered Information Security:** Safeguarding digital infrastructure requires multi-layered defensive strategies combining perimeter packet inspection firewalls, cryptographic data encryption, digital certificates, and strict statutory penalties under the Information Technology Act.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{e-Governance Models} \in \{\text{G2C (Citizens)}, \text{G2B (Businesses)}, \text{G2G (Agencies)}, \text{G2E (Employees)}\}$$

Electronic governance digital service delivery interaction models

$$\text{IPR Rights} \in \{\text{Copyright (Creative)}, \text{Patent (Inventions)}, \text{Trademark (Identity)}\}$$

Intellectual property legal protection classification

$$\text{Cyber Threats} \in \{\text{Phishing}, \text{Hacking}, \text{Denial of Service}, \text{Malware}\}$$

Taxonomy of major network and digital information threats

$$\text{Detection Rate} = \frac{\text{Identified Threats}}{\text{Total Scanned Packets}} \times 100\%$$

Firewall and intrusion detection effectiveness ratio

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Free and Open Source Software (FOSS) guarantees the freedom to run, study, modify, and redistribute source code freely.

• **Alert:** Phishing deceives users into revealing sensitive credentials, whereas malware includes Trojans, viruses, and ransomware attacking computer systems.

• **Alert:** Safe e-waste recycling neutralizes heavy metals (lead, cadmium, mercury) from contaminating groundwater ecosystems.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the applications of ICT in various sectors: e-Governance, e-Business, e-Learning, and Telemedicine. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (ICT Applications):** 1. e-Governance: Delivery of government services digitally (e.g. Akshaya centers in Kerala, passport seva). 2. e-Business or e-Commerce: Online trading, banking, digital payments (UPI). 3. e-Learning: Virtual classrooms, MOOCs (SWAYAM), digital textbooks. 4. e-Health (Telemedicine): Remote consultation and electronic medical records. **[4 Marks]**

• **Key Exam Result:** e-Governance, e-Business, e-Learning, e-Health

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2. What is Intellectual Property Right (IPR)? Explain Copyright, Patent, and Trademark. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (IPR Definition):** Legal rights granted to creators for their intellectual creations and inventions. **[1 Mark]**

• **Step 2 (Categories):** Copyright: Protects literary, musical, and software code creations. Patent: Protects new, useful, and non-obvious industrial inventions and manufacturing processes. Trademark: Protects distinctive logos, brand names, and symbols distinguishing goods (e.g. Apple logo, Nike swoosh). **[2 Marks]**

• **Key Exam Result:** Copyright (art/code), Patent (inventions), Trademark (brand logos)

$$\text{IPR Rights} \in \{\text{Copyright (Creative)}, \text{Patent (Inventions)}, \text{Trademark (Identity)}\}$$

Intellectual property legal protection classification

3. What is Plagiarism? How can it be prevented in academic and software environments? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Plagiarism Definition):** The act of presenting someone else's intellectual work, ideas, or writing as one's own without proper citation or credit. **[1 Mark]**

• **Step 2 (Prevention):** 1. Always acknowledge source and provide bibliographic citations. 2. Paraphrase properly using own words. 3. Use plagiarism detection software (Turnitin). **[1 Mark]**

• **Key Exam Result:** Plagiarism is intellectual theft; prevented by proper citation

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4. What is Free and Open Source Software (FOSS)? Explain the four essential software freedoms. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (FOSS Definition):** Software whose source code is freely available to anyone to inspect, modify, and redistribute without commercial restrictions. **[1 Mark]**

• **Step 2 (Four Essential Freedoms (FSF)):** Freedom 0: Run the program for any purpose. Freedom 1: Study how the program works and modify it. Freedom 2: Redistribute copies to help others. Freedom 3: Distribute modified versions to community. **[2 Marks]**

• **Key Exam Result:** Four Freedoms: Run, Study/Modify, Redistribute, Distribute modifications

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5. Explain software licensing models: GNU General Public License (GPL) and Creative Commons. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (GNU GPL):** Free software license guaranteeing users freedom to run, study, share, and modify software, requiring all derivative works to remain open source (copyleft). **[1 Mark]**

• **Step 2 (Creative Commons (CC)):** Flexible public copyright licenses allowing creators to grant public permissions for creative works (CC-BY: attribution required, CC-NC: non-commercial). **[1 Mark]**

• **Key Exam Result:** GNU GPL (software copyleft) & Creative Commons (content licenses)

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6. Explain common cybercrimes: Hacking, Phishing, Identity Theft, and Cyber Stalking. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Cyber Crimes):** 1. Hacking: Unauthorized access to computer systems or networks. 2. Phishing: Fraudulent emails/websites masquerading as legitimate entities to steal credentials/OTP. 3. Identity Theft: Stealing personal info (PAN, Aadhaar) to commit financial fraud. 4. Denial of Service (DoS or DDoS): Flooding server with fake traffic to make it crash and unavailable to legitimate users. **[4 Marks]**

• **Key Exam Result:** Hacking, Phishing (credential theft), Identity Theft, DDoS

$$\text{Cyber Threats} \in \{\text{Phishing}, \text{Hacking}, \text{Denial of Service}, \text{Malware}\}$$

Taxonomy of major network and digital information threats

7. Explain malicious software (Malware): Virus, Worm, Trojan Horse, Spyware, and Ransomware. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Malware Types):** 1. Virus: Malicious code that attaches to executable files and self-replicates when executed. 2. Worm: Standalone malware that self-replicates across networks without needing a host file. 3. Trojan Horse: Disguised as legitimate software but executes malicious payload secretly. 4. Spyware: Secretly monitors user keystrokes (keylogger) and browsing habits. 5. Ransomware: Encrypts user files and demands ransom money for decryption key. **[3 Marks]**

• **Key Exam Result:** Virus (needs host), Worm (network standalone), Trojan (disguised), Ransomware

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